Extension plan: redundant moments in a studentized Gaussian limit experiment

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1. Motivation and limit experiment

Consider a moment inequality problem based on a known vector-valued function $m(W, \theta) \in \mathbf{R}^d$. The null hypothesis at a fixed value of θ is

$$E[m(W, \theta)] \leq 0 \ .$$

Under the usual local asymptotic approximation, the vector of studentized sample moments converges to

$$X \sim N_d(\mu, \Omega) \ ,$$

where Ω is a positive-definite correlation matrix. The correlation matrix is part of the limit experiment and should be left unrestricted. Restrictions such as identity covariance or equicorrelation may be useful examples, but they do not capture the limit experiments generated by the moment inequality models of interest.

Partition the moment indices into two nonempty sets. Let $\mathcal{C}$ denote the *core* moments, which contain all identifying information, and let $\mathcal{R}$ denote the moments that are redundant for identification. Write

$$\mu = (\mu_{\mathcal{C}}', \mu_{\mathcal{R}}')' \ .$$

The proposed restriction on the standardized mean vector is

$$\text{for every } r \in \mathcal{R}, \quad \mu_r \leq \mu_c \text{ for some } c \in \mathcal{C} \ . \tag{1}$$

Equivalently, define

$$\Lambda = \left\{ \mu \in \mathbf{R}^d : \mu_r \leq \max_{c \in \mathcal{C}} \mu_c \text{ for every } r \in \mathcal{R} \right\} \ .$$

The restricted null and alternative are

$$\Lambda_0 = \Lambda \cap (-\infty, 0]^d \ , \qquad \Lambda_1 = \Lambda \setminus \Lambda_0 = \left\{ \mu \in \Lambda : \max_{c \in \mathcal{C}} \mu_c > 0 \right\} \ .$$

The last equality is the formal content of identification redundancy. A redundant moment cannot have a positive mean unless at least one core moment also has a positive mean. Therefore the core moments determine whether the null is violated, even though the redundant moments may still contain useful sampling information.

2. Why the restriction is imposed after studentization

Condition (1) is a restriction on the standardized local means. This point should be stated explicitly. If δ_j denotes the local drift of the unstandardized jth moment and σ_j is its standard deviation, then

$$\mu_j = \frac{\delta_j}{\sigma_j} \ .$$

An ordering of the unstandardized drifts, such as $\delta_r \leq \delta_c$, becomes

$$\sigma_r \mu_r \leq \sigma_c \mu_c \ ,$$

and does not in general imply $\mu_r \leq \mu_c$. The standardized restriction is therefore a substantive feature of the model, not a normalization that follows automatically from redundancy of the original moments. It is also the scale-invariant formulation: multiplying a moment by a positive constant changes neither its studentized statistic nor its standardized local mean.

An applied version of the paper should explain when the structure of the moment functions implies (1) after studentization. This is the link between the Gaussian experiment and the moment inequality models that motivate the analysis.

3. Feasible normals under arbitrary correlation

Let $A \subseteq \mathbf{R}^d$ be the acceptance region of the nonrandomized test

$$\phi_A(x) = I\{x \notin A\} \ .$$

Suppose that A is closed, convex, and coordinatewise lower, and that

$$P_0(X \in A) = 1 - \alpha \ .$$

For a closed convex set C, write

$$h_C(a) := \sup_{x \in C} a'x$$

for its support function. Since A is coordinatewise lower, $h_A(a) < \infty$ for a nonzero direction a only if $a \geq 0$.

The natural parameter in the Gaussian experiment is $\vartheta = \Omega^{-1}\mu$. Consequently, a supporting normal a generates the natural-parameter ray $\vartheta = \lambda a$ and the corresponding mean ray

$$\mu = \lambda \Omega a \ .$$

This suggests the feasible-normal set

$$\mathcal{F}(\Omega, \Lambda) := \{a \in \mathbf{R}^d : a \geq 0, \ \Omega a \in \Lambda\} \ . \tag{2}$$

Proposition 1 (Feasible-normal sufficiency)**.** *Let $X \sim N_d(\mu, \Omega)$, where Ω is a positive-definite correlation matrix, and let Λ_0 and Λ_1 be defined above. Suppose that A is closed, convex, and coordinatewise lower, satisfies $P_0(X \in A) = 1 - \alpha$, and*

$$a \neq 0, \quad h_A(a) < \infty \quad \Longrightarrow \quad a \in \mathcal{F}(\Omega, \Lambda) \ .$$

Then ϕ_A is α-admissible against all level-α tests, including randomized competitors.

The argument is the same Lehmann–Romano ray argument used in the bivariate note. If a is a finite supporting normal, the mean ray $\lambda\Omega a$ belongs to Λ. It cannot remain in the null: if $\Omega a \leq 0$, then

$$a'\Omega a \leq 0 \ ,$$

which contradicts positive definiteness because $a \neq 0$. Thus Ωa has a positive coordinate. Condition (1) then implies that some core coordinate is positive, so $\lambda\Omega a \in \Lambda_1$ for every $\lambda > 0$. Finally, coordinatewise lowerness makes $\mu = 0$ a least favorable null point and gives exact size there. These observations verify the ray and size conditions.

4. An admissible test that uses redundant moments

Proposition 1 gives a simple construction that works for every positive-definite correlation matrix. Fix any $c_0 \in \mathcal{C}$, and let $b \in \mathbf{R}^d$ satisfy

$$b_c = 0 \quad (c \in \mathcal{C}), \qquad b_r > 0 \quad (r \in \mathcal{R}) \ .$$

For $\varepsilon > 0$, define

$$a_\varepsilon = e_{c_0} + \varepsilon b \ .$$

At $\varepsilon = 0$,

$$(\Omega e_{c_0})_{c_0} = 1, \qquad (\Omega e_{c_0})_r = \Omega_{rc_0} < 1 \quad \text{for every } r \in \mathcal{R} \ .$$

The strict inequality follows from positive definiteness and the fact that Ω is a correlation matrix. Since $\mathcal{R}$ is finite, continuity implies that, for all sufficiently small $\varepsilon > 0$,

$$(\Omega a_\varepsilon)_r < (\Omega a_\varepsilon)_{c_0} \quad \text{for every } r \in \mathcal{R} \ .$$

Hence $a_\varepsilon \in \mathcal{F}(\Omega, \Lambda)$.

Let $z_{1-\alpha} = \Phi^{-1}(1-\alpha)$ and consider

$$\phi_\varepsilon(x) = I\left\{a_\varepsilon' x > z_{1-\alpha}\sqrt{a_\varepsilon'\Omega a_\varepsilon}\right\} \ . \tag{3}$$

This test assigns strictly positive weight to every redundant moment. Its acceptance region is a closed, convex, coordinatewise lower halfspace, and its only finite normal is a_ε. Proposition 1 therefore gives the following conclusion.

Proposition 2 (Admissible use of redundant moments)**.** *For every positive-definite correlation matrix Ω, there exists $\varepsilon > 0$ such that the test in* (3) *has exact size α and is α-admissible against all level-α tests. The test places strictly positive weight on every moment in $\mathcal{R}$.*

This result makes the paper's main point directly. Moments that are redundant for identification need not be irrelevant for inference. They can enter a test that is globally admissible in the Gaussian limit experiment, without imposing any special structure on the correlation matrix.

5. Proposed structure of the argument

The paper can make the distinction between identification and inference in three steps.

First, the existing local-power calculations compare tests based only on the core moments with tests that also use redundant moments. The crossing power curves show that dropping redundant moments is not uniformly desirable.

Second, Propositions 1 and 2 provide the decision-theoretic result in the general studentized Gaussian experiment. The main claim requires neither identity covariance nor equicorrelation: for every correlation matrix, some globally admissible test uses the redundant moments.

Third, the bivariate identity-covariance characterization explains why the manner in which redundant moments enter still matters. The short and gated-sum tests use only feasible normals and are globally admissible, whereas the long rectangle uses an infeasible vertical normal and is globally inadmissible. This sharp example prevents the general result from being misread as saying that every augmented test is well designed.

A full necessity characterization for arbitrary Ω is not needed for this argument. The next tasks are instead to verify Proposition 1 against the exact Lehmann–Romano statement, connect condition (1) to the motivating moment inequality models after studentization, and decide whether the admissible construction can be embedded in a familiar class of moment inequality statistics.

References

[1] Lehmann, E. L. and Romano, J. P. (2005). *Testing Statistical Hypotheses*. Third edition. Springer.